



AMERICAN INTERNATIONAL UNIVERSITY - BANGLADESH

Project Report
**Interactive E-Commerce Sales Analytic Dashboard for Business
Decision Making**

Course: Introduction to Data Science

Section: I

Summer, 2025-2026

Submitted by

Group - 8

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Date of Submission: 02 September 2026

TABLE OF CONTENTS

ABSTRACT	2
I. INTRODUCTION.....	2
II. LITERATURE REVIEW	3
1. Research Gap.....	4
2. Novelty and Contribution.....	4
III. PROBLEM STATEMENT	5
IV. METHODOLOGY	5
1. Dataset Description	5
2. Data Preprocessing	6
3. Feature Engineering	7
4. Exploratory Data Analysis	7
V. MACHINE LEARNING METHODOLOGY	9
1. Train-Test Split.....	9
2. Random Forest	9
3. XGBoost.....	9
4. Evaluation Metrics	10
VI. CUSTOMER SEGMENTATION	10
1. Customer Aggregation	10
2. Feature Scaling	10
3. Elbow Method	10
4. K-Means	11
VII. RESULTS AND DISCUSSION	11
1. Sales Analysis.....	11
2. Customer Behavior.....	12
3. Profit Analysis.....	13
4. Product Performance	14
VIII. MACHINE LEARNING RESULTS.....	15
1. Random Forest	15
2. XGBoost.....	16
3. Model Comparison.....	17
IX. K-MEANS RESULTS	18
X. BUSINESS INSIGHTS	19
XI. BUSINESS RECOMMENDATIONS.....	20
XII. LIMITATIONS	20
XIII. FUTURE WORK	21
XIV. CONCLUSION	22
XV. REFERENCES.....	22

ABSTRACT

E-commerce platforms generate a large amount of transaction data containing information about customers, products, sales, discounts, shipping, and orders. Proper analysis of this data can help businesses understand their sales performance and make better decisions. In this project, an e-commerce transaction dataset was analyzed using different Data Science techniques in the R programming language. The project included data preprocessing, exploratory data analysis, data visualization, machine learning, and customer segmentation.

A sample of 5,000 transaction records was prepared from the original dataset. Missing values were handled, duplicate records were checked, and new features such as profit, profit margin, month, year, and season were created. Exploratory analysis was then performed to study sales trends, customer behavior, profit, product performance, inventory, and product returns. For sales prediction, Random Forest Regression and XGBoost Regression were implemented using an 80:20 training and testing split. Random Forest achieved an RMSE of 542.768, MAE of 258.238, and R^2 of 0.581, while XGBoost achieved an RMSE of 563.959, MAE of 261.361, and R^2 of 0.530. Therefore, Random Forest performed better on the test dataset.

Customer segmentation was also performed using K-Means clustering. Four customer-related features were aggregated and standardized, and the Elbow Method was used to select three clusters. The results identified a small high-value customer group and a large lower-value customer group. Overall, the project demonstrates how Data Science can be used to convert raw e-commerce transaction data into meaningful information for business analysis and decision-making.

I. INTRODUCTION

E-commerce has become an important part of modern business, generating a large amount of data from customer orders, product purchases, prices, discounts, shipping costs, and order activities. This data contains useful information about sales performance, customer behavior, product performance, and business profitability. However, raw e-commerce data is often difficult to understand because it may contain thousands of records and many different variables. Therefore, proper data preprocessing, analysis, and visualization are needed to convert raw data into meaningful information that can support business decisions [1], [2].

Data Science provides different techniques for cleaning, analyzing, visualizing, and modeling datasets. Data visualization is an important part of this process because graphical representations can make complex information easier to understand. Line graphs, bar charts, histograms, box plots, and scatter plots can be used to identify sales trends, regional differences, high-performing products, and unusual values. Previous research on e-commerce data visualization has demonstrated how visualization can help users understand sales-related information and support business decision-making [1], [3]. Visualization describes patterns that already exist in the data. Machine learning can provide an additional level of analysis by learning patterns from historical data and using them to make predictions. Ensemble learning methods such as Random Forest and XGBoost are widely used for prediction because they can model complex relationships between input variables and target values [4], [5].

Customer behavior is another important aspect of e-commerce analysis. Customers may have different purchasing patterns depending on their order frequency, spending amount, and quantity purchased. Customer segmentation can help businesses identify groups of customers with similar characteristics and develop more targeted strategies [6]. K-Means is a commonly used clustering algorithm for grouping observations based on their similarity. In this project, K-Means clustering is applied to customer-level purchasing features, while the Elbow Method is used to determine a suitable number of clusters [7].

The main objectives of the project are to preprocess and prepare the e-commerce transaction dataset, handle missing values, check duplicate records, and perform exploratory data analysis. The project also aims to visualize sales, customer, profit, and product information; develop Random Forest and XGBoost models for sales prediction; compare their performance using RMSE, MAE, and R^2 ; and perform customer segmentation using K-Means clustering.

II. LITERATURE REVIEW

Data visualization is an important technique for understanding large and complex datasets. In e-commerce, transaction and customer data may contain many dimensions, including time, product, customer, region, price, discount, and sales. Visualization allows these dimensions to be presented in a simple and understandable form. Different charts can help users identify trends, comparisons, relationships, and unusual values in business data [1], [2].

The base paper used for this project, “*Data Visualization on E-Commerce Data*” (2023), focuses on applying visualization techniques to e-commerce data. The study demonstrates how e-commerce information can be presented through different visual representations to make business-related information easier to understand [1]. This study provides the main foundation for the visualization part of our project. Other research has also explored the use of visualization in e-commerce. Meng et al. discussed the use of data visualization technology in an e-commerce environment and showed how visualization can support the analysis and management of online business information [2]. Visual analytics has also been applied to understand online promotion strategies and customer-related information, demonstrating the value of visual analysis for e-commerce decision-making [7].

Machine learning provides an additional level of analysis by learning patterns from historical data and using them to make predictions. E-commerce applications of machine learning include sales prediction, customer behavior prediction, purchase intention prediction, and other business-related tasks [5], [8]. Random Forest is an ensemble learning algorithm that combines multiple decision trees to produce predictions. Breiman introduced Random Forest and showed that combining multiple trees can provide strong predictive performance while reducing some limitations of individual decision trees [3]. Because it can model nonlinear relationships between variables, Random Forest is suitable for structured business datasets. XGBoost is another widely used tree-based machine learning technique. It uses a gradient boosting approach in which decision trees are built sequentially to reduce prediction errors. Chen and Guestrin introduced XGBoost as a scalable tree boosting system for structured-data problems [4].

Previous studies have also investigated machine learning for e-commerce customer behavior. Li studied consumer behavior prediction using e-commerce data and demonstrated the usefulness of machine learning techniques for understanding customer-related behavior [5]. Other research has investigated online shoppers' purchase intentions using different machine learning and feature engineering techniques [8], [9].

Customer segmentation is another important area of e-commerce data analysis. Customer segmentation divides customers into groups based on similarities in their purchasing behavior. Customers may differ in their spending levels, order frequency, and quantity purchased. Identifying these groups can help businesses understand different types of customers and develop more targeted strategies [6], [10].

K-Means is one of the commonly used clustering algorithms for customer segmentation. It groups similar observations into a predefined number of clusters by minimizing the within-cluster variation. The K-Means approach was introduced by MacQueen and has become a widely used method for clustering numerical data [6]. In an e-commerce environment, customers can be grouped using features such as total sales, total orders, average purchase, and total quantity purchased.

1. Research Gap

The base paper, *Data Visualization on E-Commerce Data* (2023), mainly focuses on using data visualization techniques and Power BI to understand e-commerce information and support business decision-making. Its case study presents descriptive visualizations such as sales by year, state, and city. Although the paper discusses the potential integration of Machine Learning and AI with data visualization, predictive machine learning is not implemented as the main part of its e-commerce case study.

Based on this limitation, our project extends the existing approach by combining descriptive analysis with predictive and customer-level analytics. Random Forest and XGBoost are implemented for sales prediction and compared using RMSE, MAE, and R^2 . In addition, K-Means clustering with the Elbow Method is applied to identify different customer groups based on purchasing behavior. The project also includes detailed profit analysis, product performance evaluation, inventory-level analysis, and product return analysis. Therefore, the main research gap addressed by this project is the lack of an integrated approach that combines e-commerce visualization, machine learning-based prediction, profit analysis, product analysis, and customer segmentation within a single analytical workflow.

2. Novelty and Contribution

Based on the identified research gap, this project extends the existing visualization-focused approach by integrating descriptive, predictive, and customer-level analytics into a single e-commerce data analysis workflow. The main novelty of the project is the combination of e-commerce data visualization, Random Forest and XGBoost-based sales prediction, K-Means customer segmentation, and detailed business performance analysis.

Unlike the base study, which mainly focuses on descriptive visualization, this project implements and compares two machine learning models for sales prediction using RMSE, MAE, and R^2 . In addition, K-Means clustering with the Elbow Method is

applied to identify customer groups based on purchasing behavior. The project further incorporates profit and loss analysis, profit margin analysis, product performance, inventory-level analysis, and product return analysis.

Therefore, the contribution of this project is an integrated analytical workflow that provides both descriptive and predictive insights while also identifying customer segments and important business performance indicators. This approach provides a broader view of e-commerce data and can support more informed business decision-making.

III. PROBLEM STATEMENT

1. The existing study mainly focuses on descriptive e-commerce data visualization using Power BI.
2. It shows existing sales patterns but does not sufficiently provide predictive sales analysis.
3. Machine Learning models are discussed but are not implemented as a major part of the base study.
4. The existing approach does not include customer segmentation based on purchasing behavior.
5. Profit, product performance, inventory, and return analysis are not integrated into one analytical workflow.
6. Therefore, an integrated e-commerce analytics approach is needed to combine descriptive, predictive, and customer-level analysis.
7. This project addresses the problem by combining data visualization, Random Forest and XGBoost sales prediction, K-Means customer segmentation, profit analysis, product analysis, inventory analysis, and return analysis.

IV. METHODOLOGY

1. Dataset Description

The original dataset contained information about orders, customers, products, prices, discounts, shipping, and total sales.

Original Column	New Column
Name	Order_ID
Email	Customer_ID
Created at	Order Date
Shipping City	Region
Lineitem sku	Product ID
Lineitem name	Product_Name
Lineitem quantity	Quantity
Lineitem price	Selling_Price
Discount Amount	Discount
Shipping	Shipping Cost
Total	Sales Amount

After selecting the required columns, the dataset was reduced to 5,000 records using random sampling.

The final dataset also contained additional variables created during preprocessing and feature engineering. These included:

- Cost_Price
- Stock
- Return_Count
- Profit
- Profit_Status
- Profit_Margin
- Month
- Year
- Season

The project was implemented using the R programming language. The main R packages used were:

- readr
- dplyr
- ggplot2
- lubridate
- skimr
- corrplot
- randomForest
- xgboost
- cluster

2. Data Preprocessing

Data preprocessing was performed before the main analysis. The major steps were:

Data Import: The dataset was imported using the readr package.

Column Selection: Only the columns required for the project were selected.

Column Renaming: Long original column names were changed to shorter and easier names.

Data Sampling: The dataset was reduced to 5,000 records using: `set.seed(123)`. This seed was used to make the random sampling reproducible.

Missing Value Handling: Missing values were checked using `colSums(is.na())`.

Variable	Missing Value Treatment
Customer_ID	Guest Customer
Region	Unknown
Product_ID	Unknown SKU
Discount	0
Shipping_Cost	0
Sales_Amount	Selling Price $\times$ Quantity

Duplicate Checking: Duplicate records were checked using the `duplicated()` function.

Date Conversion: The order date was converted into date-time format using the lubridate package.

Region Standardization: Different versions of the same location name were standardized. For example:

- Dhaka, dhaka and ঢাকা were converted to Dhaka.
- Chittagong and Chattogram were standardized as Chittagong.

3. Feature Engineering

Profit: Profit was calculated as:

$$\text{Profit} = \text{Sales Amount} - (\text{Cost Price} \times \text{Quantity}) - \text{Shipping Cost}$$

Profit Status: Each transaction was classified as:

- Profit when $\text{Profit} \geq 0$
- Loss when $\text{Profit} < 0$

Profit Margin: Profit margin was calculated as:

$$\text{Profit Margin} = (\text{Profit} / \text{Sales Amount}) \times 100$$

Time Features: Three time-related variables were created:

- Month
- Year
- Season

Months	Season
March–June	Summer
July–September	Rainy
October–November	Autumn
December–February	Winter

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the main characteristics and relationships within the prepared dataset. Different statistical and visualization techniques were used to examine sales, profit, product, customer, and numerical variables. The following analyses were performed:

Sales Analysis

- Total sales
- Total orders
- Average sales
- Monthly sales
- Regional sales
- Discount analysis

Customer Analysis

- Top customers
- Regional order count
- Seasonal customer count
- Average purchase

Profit Analysis

- Total profit
- Average profit
- Monthly profit
- Regional profit
- Profit/loss status
- Profit margin distribution

Product Analysis

- Top products by sales
- Top products by profit
- Low-stock products
- Highly returned products

Statistical Visualization

A correlation heatmap was used to examine the relationships among the main numerical variables, including Quantity, Selling Price, Discount, Shipping Cost, Sales Amount, Cost Price, Stock, Return Count, Profit, and Profit Margin.

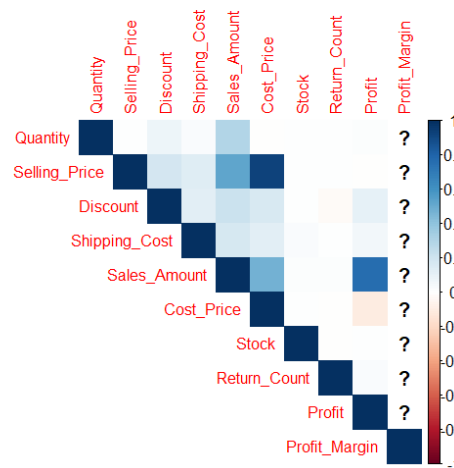


Fig. 1. Correlation Heatmap of Numerical Variables

Box plots were also used to examine the distribution and spread of Sales Amount and Profit. The box plots provide information about the median, variation, and possible outliers in the dataset. These plots are useful for identifying unusual sales or profit values that may affect the overall analysis.

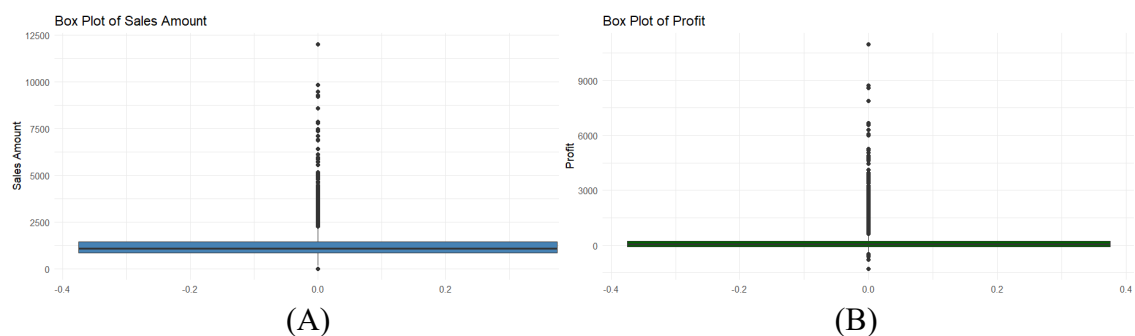


Fig. 2. Box Plots of (A) Sales Amount and (B) Profit

V. MACHINE LEARNING METHODOLOGY

1. Train-Test Split

Machine learning requires separate training and testing data. The project used an 80:20 split. The final dataset contained: 5,000 records

The data was divided into 4,000 training records and 1,000 testing records

The training data was used to build models, while the testing data was used to evaluate their performance. The target variable was Sales_Amount

The following variables were used as input features:

- Quantity
- Selling_Price
- Discount
- Shipping_Cost
- Year
- Month
- Season

2. Random Forest

Random Forest Regression was used as the first sales prediction model. It creates multiple decision trees and combines their predictions. This helps the model capture nonlinear relationships between input variables and the target variable.

The project used:

- Number of trees = 100
- Target = Sales Amount
- Training records = 4,000
- Testing records = 1,000

After training, predictions were generated for the testing dataset. The model was evaluated using RMSE, MAE, and R^2 . Feature importance was also examined to understand which variables contributed most to the prediction.

3. XGBoost

XGBoost Regression was used as the second prediction model. It is a boosting method that creates trees sequentially. Each new tree attempts to improve the errors made by previous trees. The project used the following configuration:

Parameter	Value
Boosting rounds	300
Maximum depth	6
Learning rate	0.05
Subsample	0.8
Column sampling	0.8
Objective	Regression

The model was trained using the same training dataset and evaluated using the same test dataset.

4. Evaluation Metrics

Root Mean Square Error (RMSE) measures the average size of prediction errors while giving more weight to larger errors.

$$RMSE = \sqrt{\sum (y_i - x_i)^2 / n}$$

- Lower RMSE indicates better performance.

Mean Absolute Error (MAE) represents the average absolute difference between actual and predicted values.

$$MAE = (1/n) * \sum |y_i - x_i|$$

- Lower MAE indicates better performance.

R-Squared (R²) represents how much of the variation in the target variable is explained by the model.

- Higher R² indicates better performance.

VI. CUSTOMER SEGMENTATION

1. Customer Aggregation

Customer segmentation was performed using K-Means clustering.

First, guest customers were removed from the segmentation analysis. Then, customer-level features were created by grouping transaction records by Customer_ID.

The following features were calculated:

- Total Sales
- Total Orders
- Average Sales
- Total Quantity

This transformed the transaction-level dataset into a customer-level dataset.

2. Feature Scaling

The four customer features had different numerical ranges.

For example, total sales could have a much larger numerical value than the number of orders. Therefore, feature scaling was performed using the `scale()` function.

Scaling ensures that one feature does not dominate the clustering process simply because it has larger numerical values.

3. Elbow Method

The Elbow Method was used to select a suitable number of clusters.

K values from 1 to 10 were tested. The Within-Cluster Sum of Squares (WSS) was calculated for each value of K. The results were plotted in an elbow graph.

Based on the project analysis, K = 3 was selected.

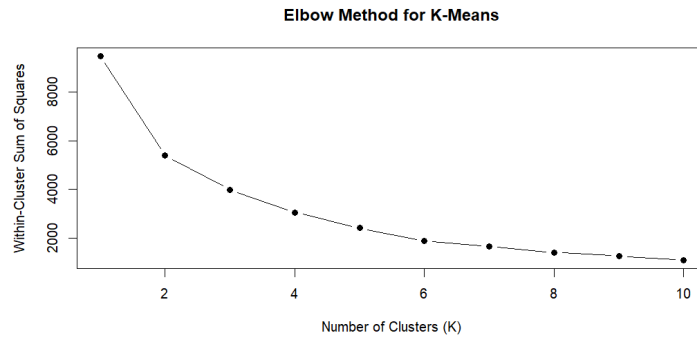


Fig. 3. Elbow Method for Selecting K

4. K-Means

K-Means clustering was then applied with Number of clusters = 3.

- Scaled Total Sales
- Scaled Total Orders
- Scaled Average Sales
- Scaled Total Quantity

Each customer was assigned to one of the three clusters.

VII. RESULTS AND DISCUSSION

1. Sales Analysis

Sales analysis was performed to understand overall sales performance. The analysis included monthly sales, regional sales, and discount-related sales behavior.

Regional Sales

The regional analysis showed that Dhaka was the highest-sales region.

Region	Total Sales
Dhaka	2,581,406
Unknown	850,478
Chittagong	202,351

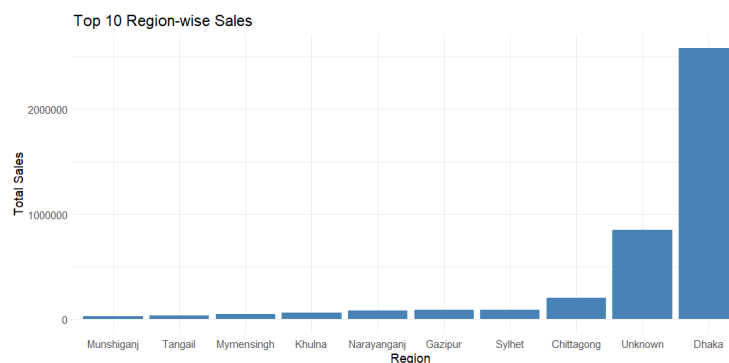


Fig. 4. Top 10 Region-Wise Sales

Monthly Sales

A monthly sales trend graph was created using a line chart. The graph helps identify changes in sales across different months and years.

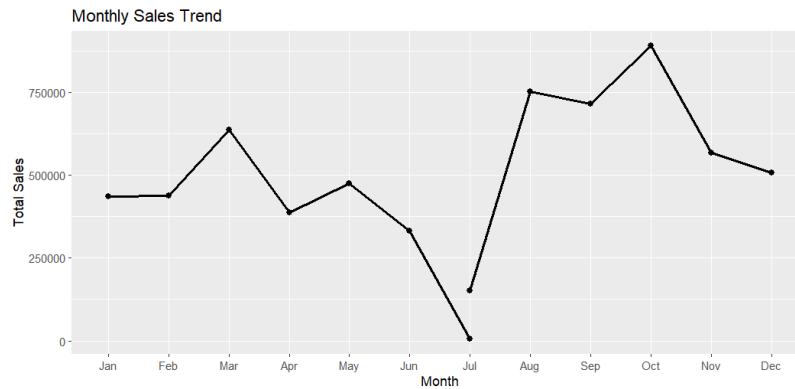


Fig. 5. Monthly Sales Trend

Discount Analysis

The relationship between discount and average sales was also analyzed. A point-and-line graph was used to compare different discount levels with average sales.

2. Customer Behavior

Customer behavior was analyzed from several perspectives. The project identified the top 10 customers based on total sales. It also compared regions based on order count.

Regional Orders

Dhaka therefore had both high sales and a high number of orders.

Region	Orders
Dhaka	1,922
Unknown	952
Chittagong	139

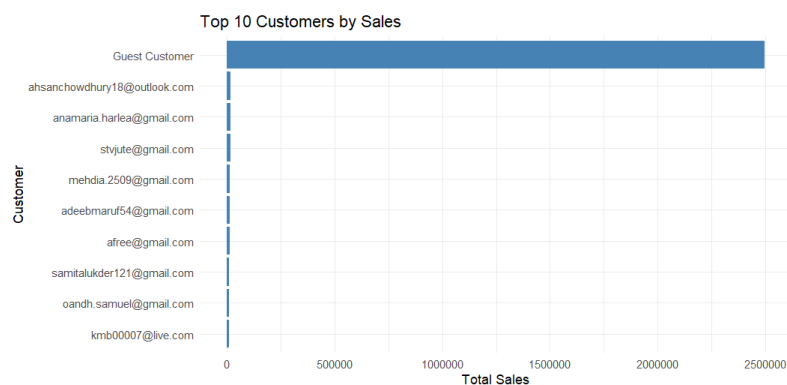


Fig. 6. Top 10 Customers by Sales

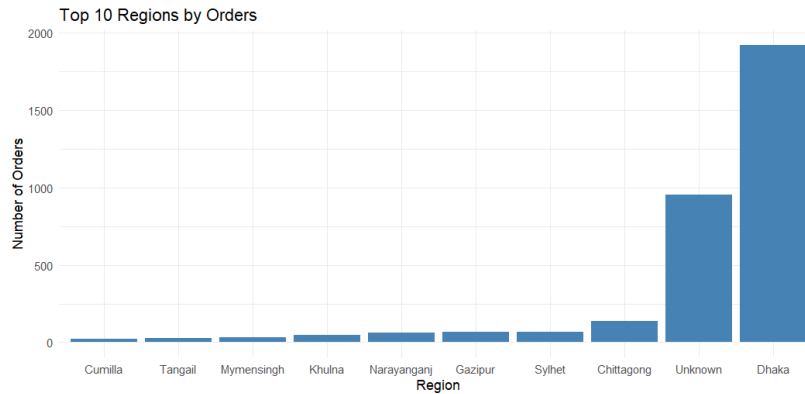


Fig. 7. Top 10 Regions by Orders

Seasonal Customers

The project also calculated the number of distinct customers in each season. This provides a basic understanding of customer activity across different periods.

Average Purchase

The average purchase value was calculated for each customer, and the top 10 customers were identified. This can be useful for identifying customers who have relatively high-value transactions.

3. Profit Analysis

Profit analysis was an important part of the project.

Measure	Result
Total Profit	963,555
Average Profit	193
Maximum Profit	10,977
Minimum Profit	-1,288
Profit Orders	3,093
Loss Orders	1,907

The total profit was positive, but the minimum profit was negative. This indicates that some transactions generated losses. The project classified transactions into two groups:

- Profit
- Loss

A total of 3,093 orders were profitable, while 1,907 orders were loss-making. The presence of loss-making orders is an important business finding. Losses may be related to the combination of selling price, cost price, quantity, discount, and shipping cost.

Therefore, businesses should not consider sales revenue alone. Profitability should also be monitored.

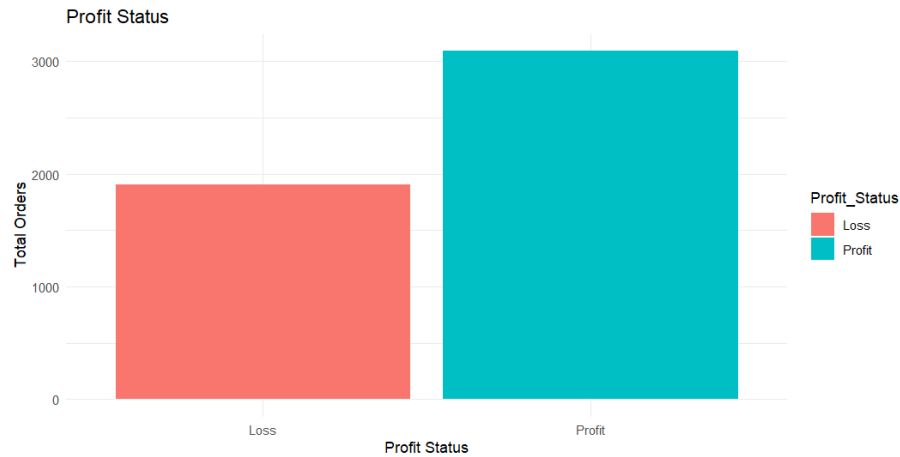


Fig. 8. Profit Status

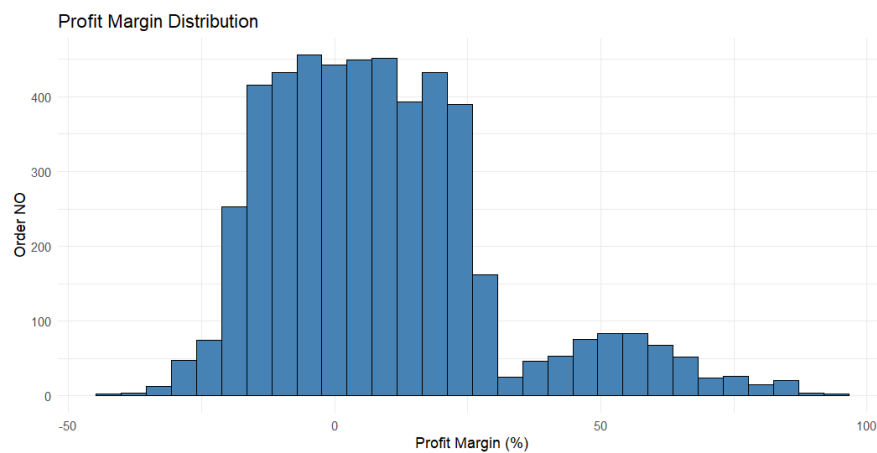


Fig. 9. Profit Margin Distribution

4. Product Performance

Product performance was analyzed using sales, profit, stock, and return information. The project identified:

- Top 10 products by sales.
- Top 10 products by profit.
- Top 10 low-stock products.
- Top 10 returned products.

Returned Products:

One important result was Christian Dior had the highest return count with 116 returns. Other highly returned products included Tomfort and MaisonVue Midnight.

This finding may indicate that these products require further investigation. Possible areas for investigation include:

- Product quality
- Customer expectations
- Delivery condition

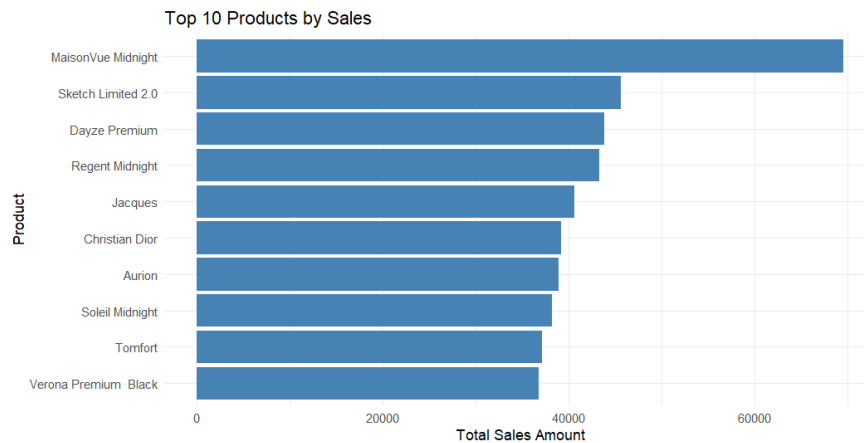


Fig. 10. Top 10 Products by Sales



Fig. 11. Top 10 Returned Products

VIII. MACHINE LEARNING RESULTS

1. Random Forest

Random Forest produced the following results:

Metric	Random Forest
RMSE	542.768
MAE	258.238
R ²	0.581

The R² value of 0.581 indicates that the model explained a meaningful portion of the variation in sales within this experimental setup. The model's feature importance analysis showed that Selling Price was the most important feature for sales prediction.

- Quantity
- Discount
- Shipping Cost
- Month
- Year

Selling price and quantity are directly related to the sales amount. Discount and shipping cost also influence the final transaction value.

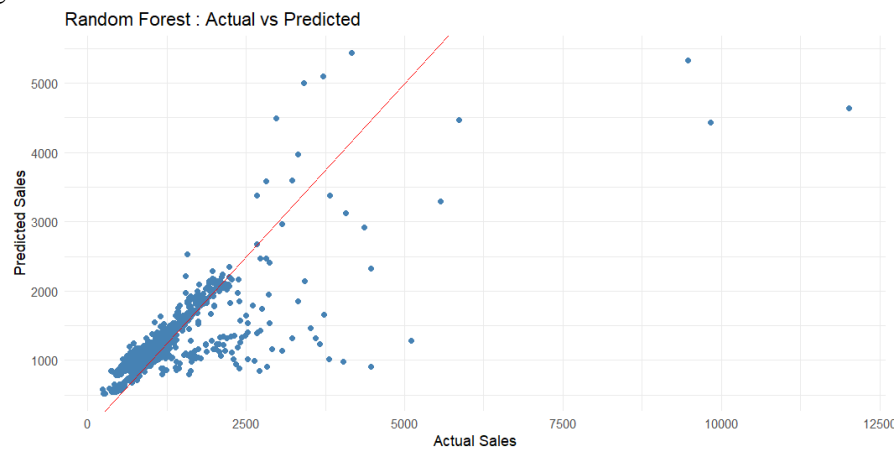


Fig. 12. Random Forest Actual vs. Predicted Sales

	IncNodePurity
Quantity	123180842
Selling_Price	589523053
Discount	326025648
Shipping_Cost	85170613
Year	7767900
Month	52138395
Season	26384817

Fig. 13. Random Forest Feature Importance

2. XGBoost

XGBoost produced the following results:

Metric	XGBoost
RMSE	563.959
MAE	261.361
R^2	0.530

The model achieved an R^2 of 0.530.

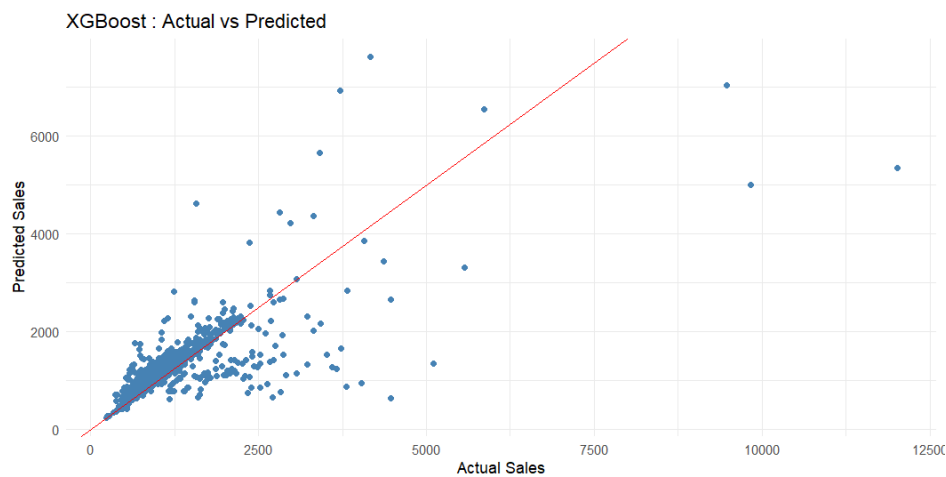


Fig. 14. XGBoost Actual vs. Predicted Sales

	Feature <char>	Gain <num>	Cover <num>	Frequency <num>
1:	Selling_Price	0.48448756693	0.417277943	0.4102053813
2:	Discount	0.23505719851	0.133900715	0.1065117020
3:	Shipping_Cost	0.09267631286	0.090937226	0.1464734915
4:	Quantity	0.07902636284	0.071592446	0.0694156981
5:	Month.L	0.02101612140	0.047928198	0.0619328132
6:	Month.C	0.01726721082	0.042414580	0.0303295654
7:	Month.Q	0.01175554322	0.021913306	0.0314440376
8:	Month^4	0.01085959315	0.015196255	0.0183887916
9:	Month^11	0.00893304093	0.027894391	0.0156822162
10:	Month^9	0.00839218747	0.032996941	0.0171947142
11:	Month^5	0.00768921630	0.021364396	0.0222894444
12:	Month^7	0.00707091044	0.024628431	0.0184683968
13:	Year	0.00427844572	0.008564149	0.0224486547
14:	Month^10	0.00340740332	0.011327614	0.0084381468
15:	Month^8	0.00294428176	0.013267364	0.0086769623
16:	Month^6	0.00283534092	0.008205098	0.0087565674
17:	SeasonSummer	0.00191475610	0.007183818	0.0021493393
18:	SeasonWinter	0.00030200901	0.001351607	0.0007164464
19:	SeasonRainy	0.00008649829	0.002055522	0.0004776310

Fig. 15. XGBoost Feature Importance

3. Model Comparison

The two models were compared using the same evaluation metrics.

Table I
Performance Comparison of Machine Learning Models

Model	RMSE ↓	MAE ↓	R ² ↑
Random Forest	542.768	258.238	0.581
XGBoost	563.959	261.361	0.530

Random Forest achieved:

- Lower RMSE than XGBoost.
- Lower MAE than XGBoost.
- Higher R² than XGBoost.

Therefore, **Random Forest performed better than XGBoost for the tested dataset and selected features.**

However, this result does not mean Random Forest is always better than XGBoost. It only indicates that Random Forest performed better under the particular dataset, features, and parameter settings used in this project.

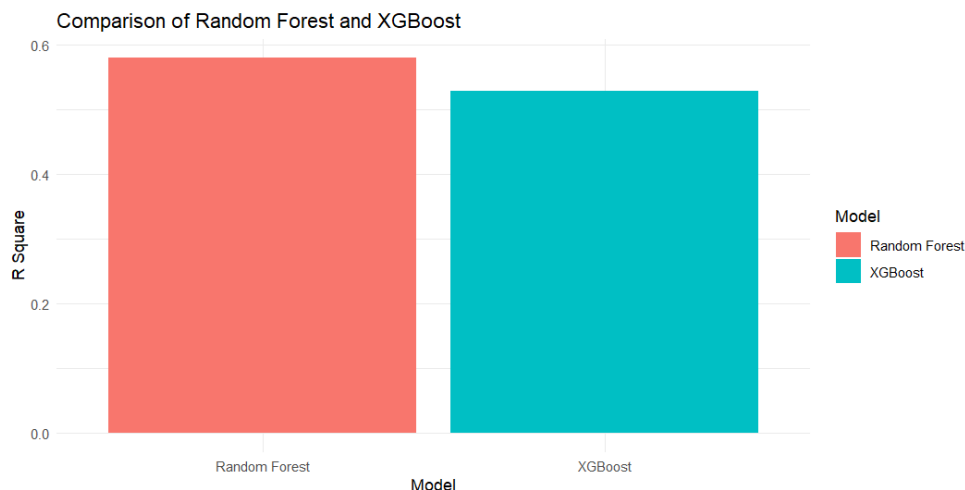


Fig. 16. Random Forest and XGBoost Performance Comparison

IX. K-MEANS RESULTS

K-Means analysis produced three customer clusters.

Table II
Customer Cluster Profile

Cluster	Customers	Average Sales	Average Orders	Average Purchase
Cluster 1	314	3,580	2.11	3,580
Cluster 2	36	8,372	5.61	8,372
Cluster 3	2,019	1,172	1.06	1,172

Cluster 1 — Regular Customers

Cluster 1 contains 314 customers.

The group has moderate average sales and order frequency.

These customers can be considered regular customers because their purchasing activity is higher than the largest low-activity group but lower than the high-value group.

Cluster 2 — High-Value Customers

Cluster 2 contains only 36 customers.

However, it has the highest:

- Average sales
- Average orders
- Average purchase

This makes Cluster 2 the most valuable customer group in terms of the selected features. Businesses could provide these customers with:

- Loyalty benefits
- Personalized offers
- Exclusive discounts
- Early access to products

Cluster 3 — Low-Value or Occasional Customers

Cluster 3 contains 2,019 customers, making it the largest group.

However, it has the lowest average sales and order frequency.

These customers may have relatively low purchasing activity.

Marketing campaigns designed to increase repeat purchases could be useful for this group.

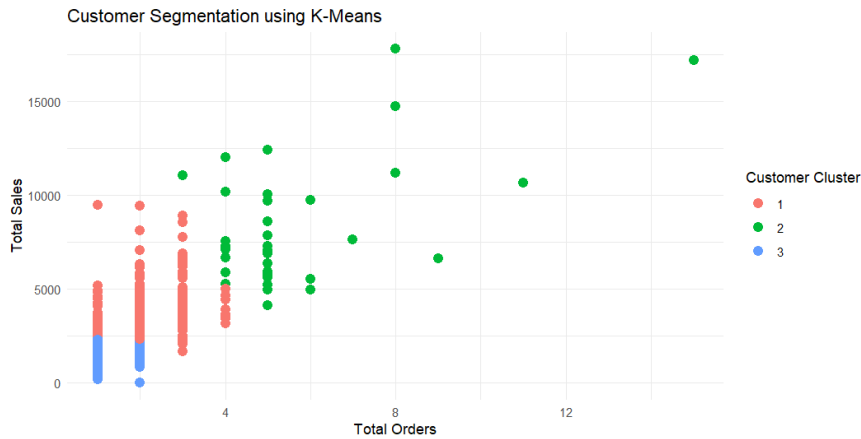


Fig. 17. Customer Segmentation Using K-Means

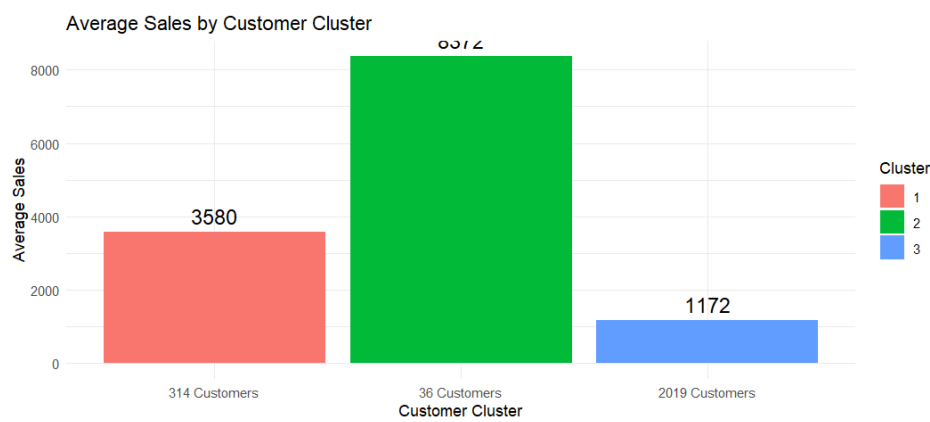


Fig. 18. Average Sales by Customer Cluster

X. BUSINESS INSIGHTS

1. Regional Sales Concentration: Dhaka generated the highest sales among the standardized regions. This indicates that regional differences are important when analyzing e-commerce performance.

2. Customer Concentration: A relatively small group of customers has considerably higher purchasing activity. The K-Means analysis identified Cluster 2 as a high-value group.

3. Profitability Problem: Although total profit was positive, 1,907 orders were classified as loss-making. Therefore, businesses should monitor individual order profitability.

4. Product Return Problem: Some products had much higher return counts than others. Christian Dior recorded the highest return count in the analyzed data.

5. Sales Prediction: Random Forest and XGBoost were able to predict sales to some degree. Random Forest performed better under the selected experimental settings.

6. Important Sales Features: Selling Price was identified as the most important feature in the Random Forest model. This suggests that price-related variables have a strong relationship with sales amount in the analyzed data.

7. Customer Segmentation: K-Means identified three customer groups with different purchasing behaviors. This can help businesses avoid treating every customer in the same way.

XI. BUSINESS RECOMMENDATIONS

1. Focus on High-Value Customers: The high-value customer group should receive personalized marketing and loyalty benefits.

2. Monitor Loss-Making Orders: Businesses should investigate orders with negative profit. Special attention should be given to:

- High discounts
- High shipping costs
- Low selling prices
- High product costs

3. Investigate Highly Returned Products: Products with unusually high return counts should be examined carefully. The business should check whether the returns are related to product quality, descriptions, delivery, or customer expectations.

4. Improve Inventory Management: Low-stock products should be monitored so that popular products do not become unavailable.

5. Use Customer Segmentation: Different marketing strategies should be developed for different customer groups. For example:

- High-value customers → Loyalty programs
- Regular customers → Personalized promotions
- Low-activity customers → Retention campaigns

6. Use Predictive Analytics: The Random Forest model tested can be used as a starting point for sales prediction. However, further model tuning and validation should be performed before using it in a real production system.

7. Improve Regional Strategy: Since sales and order activity are concentrated in certain regions, marketing and inventory strategies can be adjusted according to regional demand.

XII. LIMITATIONS

1. Dataset Size: The final analysis used 5,000 records selected from the original dataset. A larger dataset could provide more information.

2. Simulated Variables: The original dataset did not contain reliable values for all inventory-related variables. Therefore, Cost Price, Stock, and Return Count were generated for analytical demonstrations. Because of this, conclusions involving these variables should not be treated as actual business measurements.

3. Sales Prediction Approach: The project predicted transaction-level Sales_Amount rather than performing dedicated time-series forecasting of future monthly sales.

4. Limited Models Only Random Forest and XGBoost were compared. Other models such as Linear Regression, Gradient Boosting, ARIMA, SARIMA, LSTM, or GRU were not included.

5. Model Parameters: The model parameters were selected for the project and were not extensively optimized using automated hyperparameter tuning.

6. Customer Segmentation: Three clusters were selected using the Elbow Method. Other clustering methods and validation techniques could produce different groupings.

7. Data Quality: The quality of analytical results depends on the quality and completeness of the original transaction data.

XIII. FUTURE WORK

- 1. Larger Dataset:** Future research can use a larger dataset covering a longer period.
- 2. Real Inventory Data:** Actual cost price, stock level, and return information should be used instead of simulated values.
- 3. Time-Series Forecasting:** Future work can use dedicated forecasting techniques such as:
 - ARIMA
 - SARIMA
 - Prophet
 - LSTM
 - GRU
- 4. Hyperparameter Optimization:** Random Forest and XGBoost can be optimized using techniques such as grid search or random search.
- 5. Additional Machine Learning Models:** Future research can compare additional models such as:
 - Linear Regression
 - Gradient Boosting
 - Support Vector Regression
 - LightGBM
- 6. Advanced Customer Segmentation:** Future work can combine K-Means with RFM analysis using:
 - Recency
 - Frequency
 - Monetary Value
- 7. Interactive Dashboard:** The analysis can be integrated into an interactive dashboard containing:
 - Sales KPIs
 - Monthly trends
 - Regional analysis
 - Product performance
 - Customer segmentation
 - Profit monitoring
 - Sales prediction
- 8. Real-Time Analytics:** A future version could connect the system to a live e-commerce database and update the analysis automatically.

XIV. CONCLUSION

This project demonstrated how Data Science techniques can be applied to an e-commerce transaction dataset to generate meaningful business insights. The project started with data preprocessing and preparation, followed by exploratory analysis, visualization, machine learning, and customer segmentation.

A final dataset of 5,000 records was analyzed. Missing values were handled, duplicate records were checked, and additional variables such as profit, profit margin, month, year, season, stock, and return count were created. Exploratory analysis provided information about sales, customers, profits, regions, products, inventory, and returns. The machine learning part used Random Forest and XGBoost to predict sales amount. Random Forest achieved an RMSE of 542.768, MAE of 258.238, and R^2 of 0.581. XGBoost achieved an RMSE of 563.959, MAE of 261.361, and R^2 of 0.530. Based on these results, Random Forest performed better than XGBoost for the selected dataset and model settings.

The K-Means clustering analysis identified three customer segments. Cluster 2 was a small but high-value customer group, while Cluster 3 was the largest group with relatively low purchasing activity. These results demonstrate that customer segmentation can provide useful information for targeted marketing.

Overall, the project shows that combining data visualization, machine learning, and clustering can provide a more complete understanding of e-commerce data. The findings can support business decisions related to sales, customers, products, profitability, and marketing. Although the project has limitations, especially the use of simulated inventory-related variables and a limited set of machine learning models, it provides a strong practical demonstration of the Data Science workflow using R.

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